Nessma Mohdy

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Education

Bachelor of Engineering in Software and Mechatronics Engineering @ University of Calgary

- Key Courses: Embedded & Hardware Systems, Networking, Data Structures & Algorithms, Linear Algebra, Applied Operating Systems, Machine Learning, Digital Circuits, Control Systems, Computer Architecture, Software Testing, VR

Experience

Firmware Software Engineering Intern @ AMD (Markham, Ontario) May 2025 - Present

- Coordinated with various teams to build **C** x86 firmware code for servers, gaming consoles, and platform clients, streamlining integration via **CI pipelines** and rigorous **unit testing** to accelerate deployment and reliability.
- Received the **Spotlight Award** for developing an automated triage system using JIRA and Synopsys APIs in python, reducing a **2-hour** manual process to **5 minutes** and clearing **3,000+ issues** within **2** months.
- Engineered a cross-platform application with **Python and Unix** to update pre-silicon simulation software, **reducing** nightly storage use by ~**20GB** per employee per day, enabling a faster more efficient development cycle.

Machine Vision Software Engineer Intern @ Tesla (Austin, Texas) January 2025 – May 2025

- Boosted team efficiency by developing a **Flask-based** web platform with **InfluxDB** and **Caspian** to visualize cell data as heatmaps, integrating **PyTorch** and **Halcon** for advanced analysis of millions of cell images **(FTP)**.
- Initiated a comprehensive v**ision system** (camera selection, setup, installation) on cathode production preventing **4 days** of downtime and collateral loss of **5M** USD by utilizing **ResNet** for defect detection and machine vision for crack identification achieving **94+%** detection accuracy.
  - Performed **cross-validation** on **PyTorch pre-trained** models (ResNet, Vision Transformer), **YOLO**, along with multi-layer ResNet variant built from scratch leveraging **CUDA** for GPU acceleration.
- Optimized **Optical Character Recognition** for cell IDs, reducing recognition time to under **0.2** ms per unit and increasing reliability to **98.7%**, enabling accurate tracking and processing of **1.4 million** battery cells per week.

Controls Engineering Team Member @ Schulich Space Rover October 2023 – December 2024

- Transformed **Python-based** code to **C++** for rover control while integrating inter-module **WebSocket** communication for more structured and scalable execution.
- Implemented **control algorithms** to regulate motor speeds through **PID feedback loops** and **OP-AMPS** communicating through a **VINT hub** in **Python**. Significantly reducing the tipping of the rover on textured and hill-like platforms increasing the overall stability of the rover cutting recovery time by **70%**.

Data Analyst Researcher @ University of Calgary May – August 2023

- Applied non-linear supervised methods, including KNN and clustering, in **MATLAB** to achieve **83%** prediction accuracy, enabling key insights that contributed to a high-impact research publication on long COVID effects.

Projects

Rycho | *typescript, mongoDB, SQL, rest-APIs, React*

- Led the full **SDLC** for a full-stack utilizing **React** and **typescript** to create a social media application, designing relational databases hosted on **MongoDB** and integrating **Spotify's API** to support concurrent **API** calls with an average response time of less than **100ms** and **20+** users.

2048 Solver | *Selenium, Gymnasium, PyTorch, Numpy*

- **Engineered a full-stack Reinforcement Learning (RL) system** using **Chrome/Selenium** for live 2048 automation, a custom **Gymnasium** wrapper, and **PyTorch** for model training. Benchmarked RL paradigms (DQN, PPO, MCTS), resulting in an **AlphaZero-style MCTS** agent that achieved a **2046-tile.**

LongTimeNoCrypto | *C++, Networking*

- Customized SHA-256 (achieving <30 blocks/sec) and designed a proprietary Cipher Hill algorithm to generate **cryptographic hash blocks**. Engineered a **peer-to-peer** messaging system embedding encrypted messages into a custom **blockchain** for secure transmission, demonstrating sub-50ms message propagation across 5+ nodes.

Eye Controlled Robotic Hand | *OpenCV, ROS, Gazebo, Python*

- Developed a real-time control system that utilizes **OpenCV** eye-tracking with **ROS** to manipulate a simulated dexterous robotic hand in **Gazebo**. Applied linear algebra and trigonometry for **forward kinematics**, and applied **reinforcement learning (TorchRL)** to optimize hand movement precision and responsiveness.

Technical Skills

Coding Languages: C/C++, Python, Java, SQL, Arduino, JavaScript, HTML, MATLAB, C#, ROS, Assembly
Platforms: Linux/Unix (WSL), Windows, Mac OS
Tools: Git, CAD, Unity, VS Code, Blender, MySQL, Quartus, MPLAB X, XML, UTM, TCP communication, Microcontrollers, RTOS, UART, GUI, Jira, Docker, Terraform, Ansible, Kubernetes, Excel, Agile, Eclipse, Simulink